Asya Ünal

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Research interests: robot learning, diffusion policies and planning, VLA models, robot perception for manufacturing

EDUCATION

Bilkent University, B.S. Computer Science — *Sep 2023 – Jun 2027 (expected)*
GPA 3.65/4.00 · 60% Merit Scholarship · All departmental coursework completed
City University of Hong Kong, CS Exchange (Computer Vision, AI, Signals & Systems) — *Jan – May 2026*
Ankara University, Faculty of Medicine, three years completed before switching to CS — *Sep 2020 – Sep 2023*

RESEARCH EXPERIENCE

Bilkent University, LiRA Lab | *Undergraduate Researcher, Asst. Prof. Özgür S. Öğüz* — *Sep 2026 – Present*

- Built DP-C (convolutional, inpainting) and DP-T (Transformer, cross-attention) **DDPM diffusion policies** for simulated **Push-T** from scratch in PyTorch: a reduced-budget keypoint reproduction of Chi et al. [code]
- Trained 6 runs (2 architectures × 3 seeds) to mean best-checkpoint target coverage **0.995** (DP-C) / **0.991** (DP-T); reproduced stability, action-horizon and observation-delay ablations, finding lower action error did not guarantee success
- Implemented **Diffuser** (Janner et al.) as a **receding-horizon trajectory planner** on D3IL Avoiding in PyTorch/NumPy only, ablating generation horizon $G \in \{4, 32, 64\}$ against action horizon $\{1, G/2, G-1\}$

University of Maryland | *Undergraduate Researcher, Prof. A. Yavuz Oruç* — *Oct 2026 – Present*

- Researching **when a frozen vision-language-action (VLA) policy should replan**, and whether adaptive replanning beats tuned fixed schedules at equal compute (**SmolVLA** on MetaWorld and LIBERO)
- Integrating provenance tracking into preregistered tests with exact replay, counterfactual rollouts and query-cost accounting

KUIS AI Center, Koç University | *Undergraduate Researcher, Prof. A. Murat Tekalp* — *Sep 2026 – Present*

- Investigating **semantic fidelity in diffusion super-resolution**: using OCR and VLMs to test whether text reconstructed by **DiffTSR** on **TextZoom** is grounded in the low-resolution input
- Fine-tuned **Qwen2.5-VL-3B** with **LoRA** as a reference-free judge of reconstruction candidates; engineered resumable train/eval pipelines with source-grouped splits and strict vs. diagnostic scoring

PUBLICATIONS

- **A. Ünal** et al., "AI-Driven Collaborative Assembly Line Inspection: System Integration and Deployment Challenges," *MobiWIS 2026*, Granada. (Accepted)
- **A. Ünal** et al., "Surface and Edge Defect Inspections on Metals Using Deep Learning and Optical Camera Data," *IEEE FiCloud 2025*, Istanbul.
- F. Kaya, **A. Ünal** et al., "Real-Time Anomaly Detection in Industry 4.0 Using Asset Administration Shell," *Springer LNCS 16066*, MobiWIS 2025.

WORK EXPERIENCE

CodeAPeel | *Software & Product Developer, with Prof. A. Yavuz Oruç (UMD)* — *Aug 2026 – Present*

- Built a **FastAPI** AI assistant around an **execute–observe–repair** loop: generated assembly runs in a bounded sandbox and is verified before any editor change is proposed
- Lead Java desktop tools and a **React/TypeScript** workspace for MIPS, RISC-V, VESP and Ark-U architectures

Mobile Action | *Software Engineering Intern* — *Jun – Jul 2025*

- Initiated and built a **MCP server** exposing App Store analytics to LLM agents; shipped SearchAds features (Vue 3, Spring Boot, RabbitMQ)

Eluvium, London | *AI Engineer Intern* — *May – Jun 2025*

- Rebuilt a monolithic state machine as a modular **LangGraph** workflow; built a LangChain manufacturing-alert router

Teknopar Industrial Automation | *Machine Learning Intern* — *Jul – Aug 2024*

- Trained CNN and **YOLO** models for metal surface and edge defect inspection from optical camera data: **98% test accuracy**
- Fine-tuned **DistilBERT** to classify 3K+ imbalanced industrial error logs at **85% accuracy**

SELECTED PROJECTS & LEADERSHIP

- **Miper** [miper.ai], Founder: multimodal (voice, photo, text) memory system on FastAPI, LangGraph, Whisper and pgvector
- **Qardex** [qardex.app], Founder: AI study app that turns notes into adaptive flashcards and live multiplayer races
- **ACM Bilkent**, President (2024–25) and Head of Audit Committee (2025–26) of a 400+ member chapter
- **Bilkent CS Class Representative** (2026–27): voice of my year to the department and student member of its **Quality Management in Education (QME)** team, turning peer feedback into action items

SKILLS

Robot learning: diffusion policies (DDPM), imitation learning, receding-horizon planning, Push-T, D3IL, PyTorch, NumPy
Perception & ML: computer vision (CNNs, YOLO), VLMs, LoRA, Hugging Face, diffusion models, LLM agents, LangGraph
Programming & Systems: Python, C++, Java, TypeScript, C, FastAPI, React, PostgreSQL/pgvector, Docker
Languages: Turkish (native), English (fluent, IELTS 8.0/9), German (B1)